UNIT-4

Choose the best answer:

1. b) in case of Sc, 3d orbital are partially filled but in Zn these are completely filled
2. a) Cr

   $Cr \Rightarrow [Ar]3d^5 4s^1$
3. a) Ti
4. c) Ni^{2+}
5. a) 5.92BM

   $Mn^{2+} \Rightarrow 3d^5$contains 5 unpaired electrons

   n=5 ; $\therefore = \sqrt{n(n+2)}BM$

   $\therefore = \sqrt{5(5+2)} = \sqrt{35} = 5.92BM$
6. b) Ti^{4+}

   Ti^{4+} contains no unpaired electrons in d orbital, hence no d-d transition.
7. c) their ability to adopt variable oxidation states
8. a) $VO_2^+ < Cr_2O_7^{2-} < MnO_4^-$

   $\overset{+5}{V}O_2^+ < \overset{+6}{Cr}_2O_7^{2-} < \overset{+7}{Mn}O_4^-$ greater the oxidation state, higher is the oxidising power
9. d) brass

   Brass contains 70% Cu + 30% Zn
10. b) $(NH_4)_2Cr_2O_7$

    $2K_2Cr_2O_7 \longrightarrow 2K_2CrO_4 + Cr_2O_3 + \left(\frac{3}{2}\right)O_2$

    $(NH_4)_2Cr_2O_7 \longrightarrow Cr_2O_3 + 4H_2O + N_2$

    $KClO_3 \longrightarrow KCl + \left(\frac{3}{2}\right)O_2$

    $Zn(ClO_3)_2 \longrightarrow ZnCl_2 + 3O_2$
11. b) Carbon dioxide

    $5(COO)_2^{2-} + 2MnO_4^- + 16H^+ \longrightarrow 2Mn^{2+} + 10CO_2 \uparrow + 8H_2O$
12. b) $Na_2Cr_2O_7$ is preferred over $K_2Cr_2O_7$ in volumetric analysis
13. b) Mn^{2+}

    $MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$
14. a) Na_2SO_3 , SO_2 , $Cr_2(SO_4)_3$

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